Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 34

Verifiable Random Function (VRF)-II

Estimated Reading Time: 15–20 minutes

Introduction

Verifiable Random Function – II focuses on the practical role of VRFs in blockchain protocols, especially validator selection, cryptographic verification, and decentralized consensus.

In a blockchain network, selecting participants unpredictably is important for reducing the possibility of targeted attacks and manipulation. A VRF provides a cryptographic mechanism for generating verifiable outputs that can be used as part of a selection process.

VRF-Based Validator Selection

A blockchain network may need to select a small group of participants to propose or validate a block.

If selection is predictable, malicious participants may target the selected validators. If selection is controlled by a single authority, that authority may manipulate the process.

A VRF-based selection mechanism can allow participants to independently calculate an output and provide proof of their eligibility.

A typical process is:

1. The blockchain protocol defines an input for a particular round or selection event.
2. Each eligible participant evaluates the VRF using their private key.
3. The participant obtains a VRF output and proof.
4. The protocol applies a predefined eligibility rule to the output.
5. Eligible participants submit the required proof.
6. Other nodes verify the proof and determine whether the participant satisfies the selection conditions.

The exact process varies between blockchain protocols.

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Example of VRF-Based Selection

Suppose a blockchain network has 100 eligible participants and needs to select a small committee for a consensus round.

Each participant evaluates a VRF using a protocol-defined input and their private key. The resulting outputs are compared against a predefined eligibility threshold.

Participants whose outputs satisfy the eligibility condition can submit their proofs. Other nodes verify those proofs and check whether the selection rule has been followed.

The VRF helps make eligibility verifiable without requiring participants to trust a central authority to announce random selections.

The actual probability of selection depends on the protocol's design, such as participant stake, eligibility thresholds, and other rules.

Role of VRFs in Blockchain Consensus

VRFs can support consensus mechanisms in several ways:

- Unpredictable selection: Helps make it difficult to predict which eligible participant will be selected.
- Public verification: Allows nodes to verify eligibility proofs.
- Reduced reliance on centralized randomness: Participants can generate and verify outputs without trusting a single random-number provider.
- Support for scalable protocols: Some protocols use VRF-based selection to identify a smaller subset of participants for particular consensus tasks.

A VRF does not replace the entire consensus mechanism. It is a cryptographic component used within a broader protocol.

VRF and Digital Signatures: Difference

Although VRFs and digital signatures both use public-key cryptography, their purposes differ.

Feature	Digital Signature	Verifiable Random Function
Main purpose	Authenticate signed data	Generate verifiable pseudorandom output
Private-key use	Creates a signature	Generates an output and proof

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Public-key use	Verifies a signature	Verifies the VRF output and proof
Typical blockchain application	Transaction authorization	Randomized validator selection
Main result	Signature	Output and proof

A digital signature establishes that a signature is valid for particular data and a public key. A VRF establishes that a particular output is valid for an input and public key under the VRF's security properties.

VRF and Ordinary Random Number Generation

Ordinary random-number generation produces random values, but other participants may not be able to verify how those values were obtained.

A VRF produces an output together with a cryptographic proof. This allows others to verify the output without knowing the private key.

However, a VRF output is not necessarily unpredictable to its key holder. A participant knows their own private key and can calculate the output for a known input. Therefore, blockchain protocols must carefully design their inputs and selection rules to limit opportunities for manipulation.

Security Considerations and Limitations

Important considerations include:

- Private keys must be protected.
- Public keys must be correctly associated with participants.
- Inputs should be defined by the protocol to prevent strategic manipulation.
- Eligibility thresholds and selection rules must be designed carefully.
- Proofs must be verified correctly.
- VRFs do not independently prevent denial-of-service attacks, collusion, or all forms of consensus manipulation.

Security depends on the VRF construction, implementation, and the surrounding blockchain protocol.

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Applications and Examples

VRFs are relevant to blockchain systems that require verifiable, unpredictable participant selection. A prominent example is Algorand, which uses VRF-based cryptographic sortition in its consensus protocol.

Other possible applications include randomized committee formation, verifiable lotteries, and protocol-specific assignment of tasks.

Summary

VRF – II explains how verifiable randomness can be integrated into blockchain consensus. Participants generate VRF outputs and proofs, while other nodes verify the results and apply protocol-defined eligibility rules. VRFs support verifiable participant selection, but their security and fairness depend on correct implementation and the design of the overall blockchain protocol.

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